

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 Total Marks:50

Annexure

Constraint - Based Problem-Solving

Duration: 2Hrs

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.
2. Implement an ϵ -greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ϵ values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.
3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints.
4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.

5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.
6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.
7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth. Explain how bandwidth constraints influence the reward function and policy optimization.
8. Analyse an experimental comparison between random action selection and ϵ -greedy action selection in a Multi-Armed Bandit problem under a fixed time constraint. Record cumulative rewards and explain which strategy provides better performance.
9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.
10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

Code of Conduct:

I (M.S.Dinesh / 192424016) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. S. Dinesh

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

AI3

i) Question 1: Theoretical framework of Reinforcement learning in Grid World

Reinforcement learning (RL) models required sequential decision making problems where an agent interacts with an environment to learn an optimal behaviour policy.

i) State Space (S)

The state space encompasses all distinct configurations that agent can occupy in the environment.

Definition: A discrete set of two dimensional coordinate pair $S = \{(x, y) \mid x \in \{0, 1, \dots, N-1\}, y \in \{0, 1, \dots, M-1\}\}$, where $N \times M$ is grid resolution.

• Terminal states: Subsets S_{goal} and $S_{obstacle}$ terminate an episode upon entry.

ii) Action Space (A)

The action space defines the set of allowable moves available to the agent at any non-terminal state s .

Definition: A discrete, finite set of directional transitions $A = \{\text{North, South, East, West}\}$.

Dynamics: Action a on a state $s = (x, y)$ to a candidate next state $s' = (x', y')$.

reward function (R)

The scalar feedback signal $R(s, a, s')$ for policy development by evaluating state-action transitions

- Goal state ($s' \in S_{goal}$): High positive scalar (+100) to incentivize reaching the destination

- Obstacle state ($s' \in S_{obstacle}$): large negative penalty (-10) to enforce safety constraints

- Step penalty ($s' \in S_{step}$): Small negative cost (-1) per move to penalize inefficient, loitering paths

iv) constraints & environment

- Boundary constraints: Hard spatial limit that reject actions moving the agent beyond grid indices ($0 \leq x < N$, $0 \leq y < M$)

2.) Theory of Multi Armed Bandits & Greedy Strategy

The Multi Armed Bandit (MAB) Problem models the fundamental exploration vs. exploitation dilemma in the non associative environment

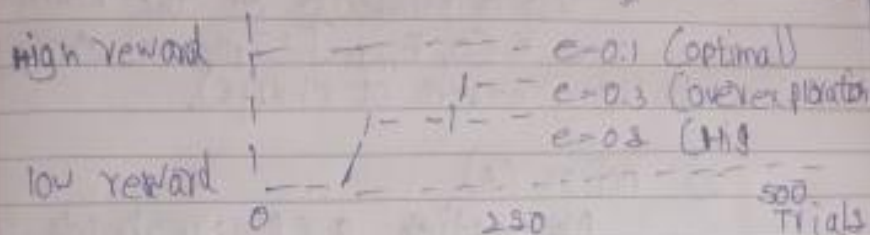
Mathematical function

ie discrete choices, arm $a \in \{1, \dots, K\}$ yield a reward mean expected value
$$v_a(a) = E[R_t | a_t] = a$$

$$Q_{\pi}(a) = \frac{\sum_{i=1}^T R_i \cdot \mathbb{I}(A_i = a)}{\sum_{i=1}^T \mathbb{I}(A_i = a)}$$

ϵ - Greedy Action selection

$$A_t = \begin{cases} \arg \max_a Q_t(a) & \text{with probability } (1 - \epsilon) \\ \text{a random arm } a \in A & \text{with probability } \epsilon \end{cases}$$



i) $\epsilon = 0.1$ (low Exploration / High exploitation)

Exploration: 10% of trial explore non greedy arms

Exploitation: 90% of trial select the current best estimate

ii) $\epsilon = 0.3$ (Moderate Exploration)

Exploration: 30% of trials are gather data across all arm

Exploitation: 70% of trial exploit current knowledge

Q3 Markov Decision process MDP for energy constrained Autonomous Robots

i) states (s_t)

A state $s \in S$ represents a complete Snapshot of the robot physical and internal condition $(s; A, P, R, E)$

1.) states (s)

- A state $s \in S$ represent a complete Snapshot of robot position coordinate $(x, y) \in \mathbb{R}^2$
- $v \geq 0$: velocity of the robot
- $e \in (0, E_{max}]$: Internal energy (battery state of charge).

2.) action (A)

An action $a \in A$ corresponds to continuous control command send to a ctuator

- $a = (a_{accel}, \theta)$
- $a_{accel} \in [-a_{max}, a_{max}]$: linear acceleration
- $\theta \in [-\pi, \pi]$

3.) transition probabilities (p)

$P(s' | s, a)$ represent the stochastic probability density of reaching s'

• kinematic techniques

$$x' = x + v \cos(\theta) \Delta t + \epsilon_x, y' = y + v \sin(\theta) \Delta t + \epsilon_y$$

• energy Depletion

$$e' = e \frac{P_{base line} + k_1 \cdot v^2 + k_2 \cdot |a_{accel}|}{D_e(v, a)}$$

$$R(s, a, s') = \begin{cases} \text{destination} \\ -w_e \cdot D_e(v, a) - \frac{1}{2} \cdot D_e \\ \text{reward} \end{cases}$$

Q4 value iteration theory and Bellman optimality equations

i) Bellman expectation equation for states values

$$V^\pi(s) = \sum_{a \in A} \pi(a|s) \sum_{s' \in S} P(s'|s, a) [R(s, a, s') + \gamma V^\pi(s')]$$

ii) Bellman optimality equation

$$V^*(s) = \max_{a \in A} \sum_{s' \in S} P(s'|s, a) [R(s, a, s') + \gamma V^*(s')]$$

iii) how iterative value works

$$V^{(k+1)}(s) \leftarrow \max_{a \in A} \sum_{s' \in S} P(s'|s, a) [R(s, a, s') + \gamma V^{(k)}(s')] \quad \forall s \in S$$

iv) handling constraints / restricted states

s, s' restricted are handled by forcing their transition probability to zero, setting their value permanently to $V(s) = -\infty$ or imposing infinite penalties so surrounding state assign near-zero probability to moving forward there

$$\max_s [V^{(k+1)}(s) - V^{(k)}(s)] < \epsilon$$

the deterministic optimal policy $\pi^*(s)$ is extracted by taking the greedy action

$$\pi^*(s) = \arg \max_{a \in A} \sum_{s'} (P(s')/s(a)) [R(s, s') + \gamma V^*(s')]$$

Q6) Constrained RL Framework for Warehouse Robot Deliveries

i) framework Architecture

Environment Warehouse
Grid, Package station, Recharge

action a_t
(move, drop)

state s_t

Agent

policy π | $\pi_t = \arg \max_{a \in A} (Q(s_t, a))$
(log probability)

Dual Optimizer (Lagrange Multiplier)
 λ

ii) formula & Design specification

$s_t = (x, y, b, c, d)$
Robot location, carried package, activate pending delivery

e_t Battery (0-100%), Remaining

Date _____
Action Space (A)

$a_t \in \{\text{Navigate } (x, y), \text{Pick Package, Drop package, Dock at charger}\}$

constraint Design

$$\max_{\pi} E_{\pi} \left[\sum_{t=0}^T R_{\text{delivery}}(s_t, a_t) \right]$$

$$E_{\pi} \left[\sum_{t=0}^T c(s_t, a_t) \right] \leq E_{\text{max}}$$

i) Reward Term R_{delivery}

- +100 for completing a successful
- -1 per unit time step

ii) Cost penalty term $c(s, a)$:

quantifies real-time battery per usage